EXPERIMENT : 9

Objectives: To create a RESTful API for managing student records using Django REST Framework.

Tools Required:

Visual Studio Code

Web Browser

Python 3.x

Django Framework

Django REST Framework [DRF]

Procedure:

1. Create Django project api
2. Create app students and add to settings
3. Install and add Django REST Framework
4. Create Student model (name, pr)
5. Apply migrations
6. Register model in admin
7. Create serializer for Students
8. Create API views (CRUD operations)
9. Configure URLs (/api/students/)
10. Run server and test API

Commands Used:

1. pip install django djangorestframework.
2. django-admin startproject myproject.
3. cd myproject.
4. python manage.py startapp studentapi.
5. python manage.py makemigrations.

6. python manage.py migrate.

7. python manage.py runserver.

models.py

myproject > Studentapi > models.py > Student > __str__

```python
from django.db import models

class Student(models.Model):
    name = models.CharField(max_length=100)
    age = models.IntegerField()
    branch = models.CharField(max_length=100)
    email = models.EmailField()
    phone = models.CharField(max_length=15)

    def __str__(self):
        return self.name
```

Serializers.py

myproject > Studentapi > serializers.py > StudentSerializer > Meta

```python
from rest_framework import serializers
from .models import Student

class StudentSerializer(serializers.ModelSerializer):
    class Meta:
        model = Student
        fields = '__all__'
```

Urls.py

myproject > myproject > urls.py > ...

```python
from django.contrib import admin
from django.urls import path, include

urlpatterns = [
    path('admin/', admin.site.urls),
    path('student/', include('Studentapi.urls')),
    path('api/', include('api.urls')),
    path('myapp/', include('myapp.urls')),
      path('register/', include('registrationapp.urls')),
      path('blog/', include('blog.urls')),
       path('accounts/', include('accounts.urls')),
       path('', include('mainapp.urls')),
]
```

Django REST framework

Student List Create

Student List Create

OPTIONS GET

List all students or create a new student. GET /api/students/ - List all students POST /api/students/ - Create new student

```
POST /api/students/
```

```
HTTP 201 Created
Allow: GET, POST, HEAD, OPTIONS
Content-Type: application/json
Vary: Accept
```


```
{
    "id": 1,
    "name": "ABHIMANYU KUMAR",
    "roll_number": "2024071002",
    "email": "luckypal5002@gmail.com",
    "created_at": "2026-04-10T04:19:42.197074Z",
    "updated_at": "2026-04-10T04:19:42.197135Z"
}
```


Raw data | HTML form

Name ABHIMANYU KUMAR

Roll number 2024071002

Email